

SEGMENTED PRICING

Overview of segmented pricing

2

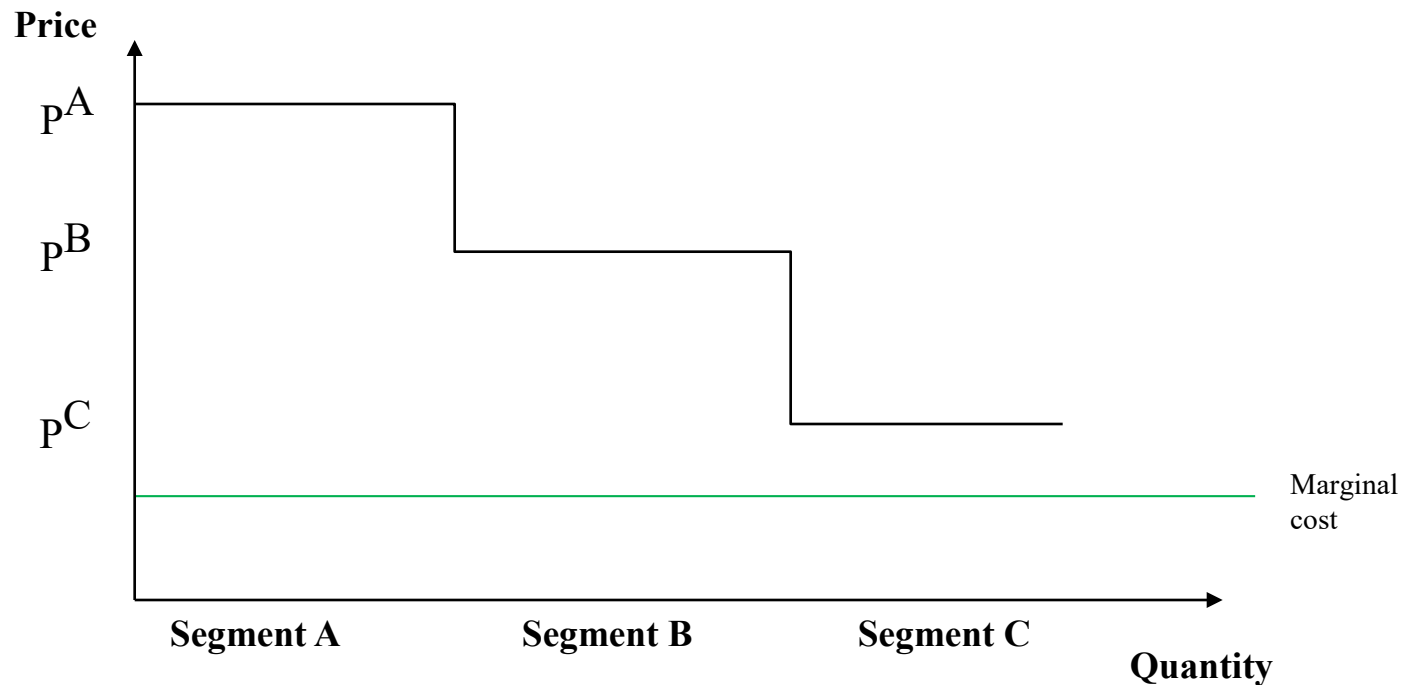
Pricing structures that raise profits by segmenting on price to capture value from customers with different tastes

- We will look at *theory* and examples
- **Targeted** (*Personalized*) **Pricing**
- **Product Line Pricing Strategy** as indirect segmented pricing
- Product Line Pricing tactics & **cannibalization**

Recall: Optimal Uniform Price

Charge customers same price for same product

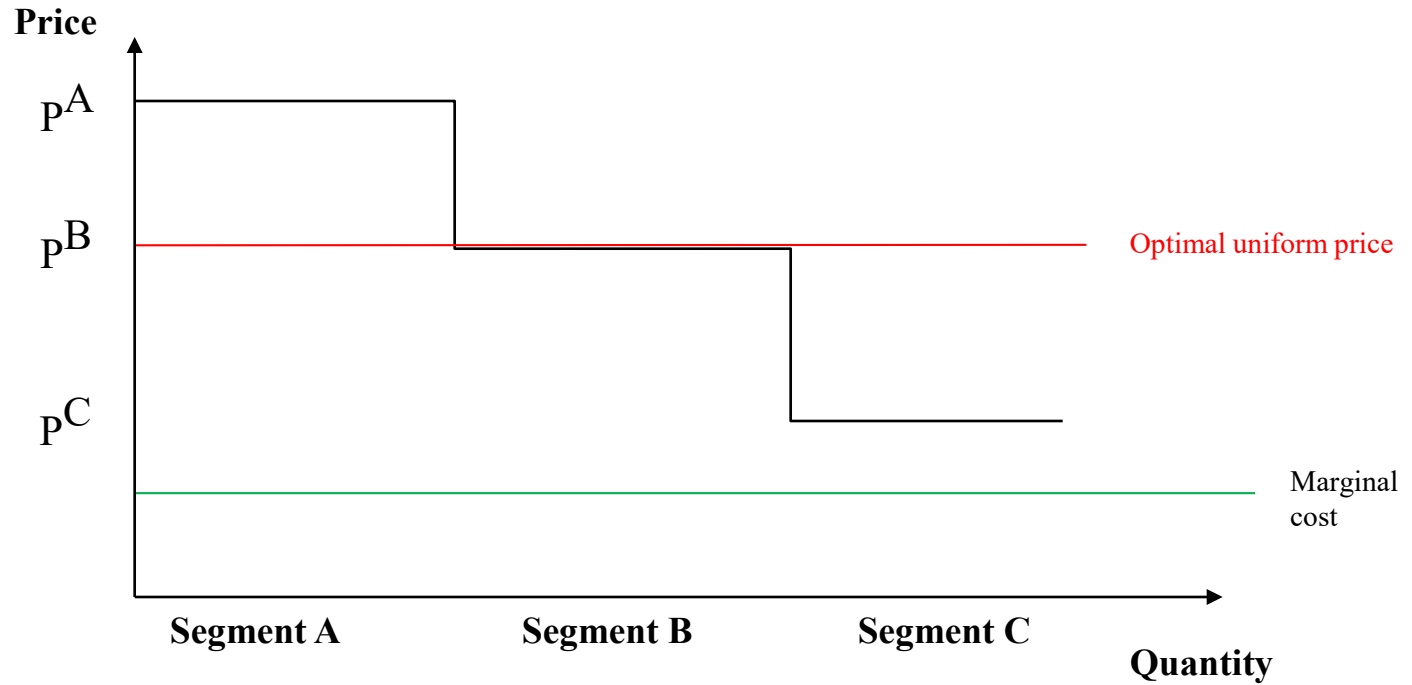
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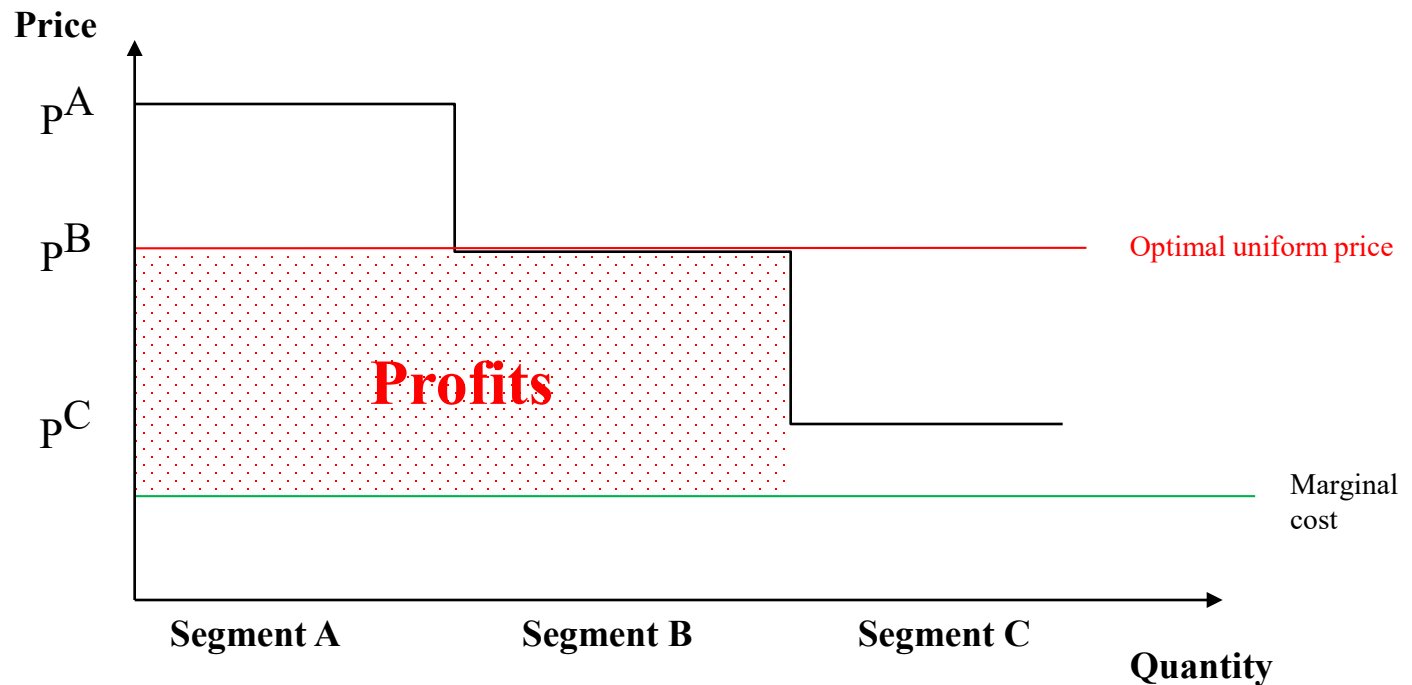
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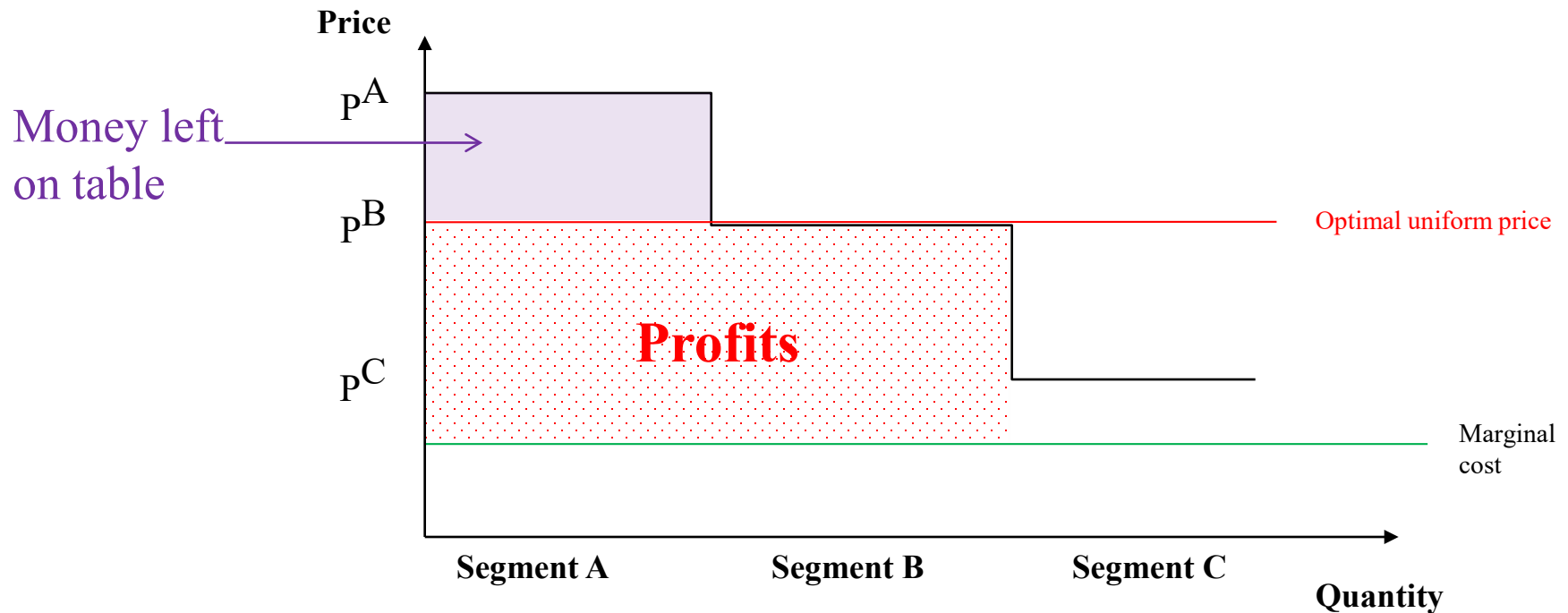
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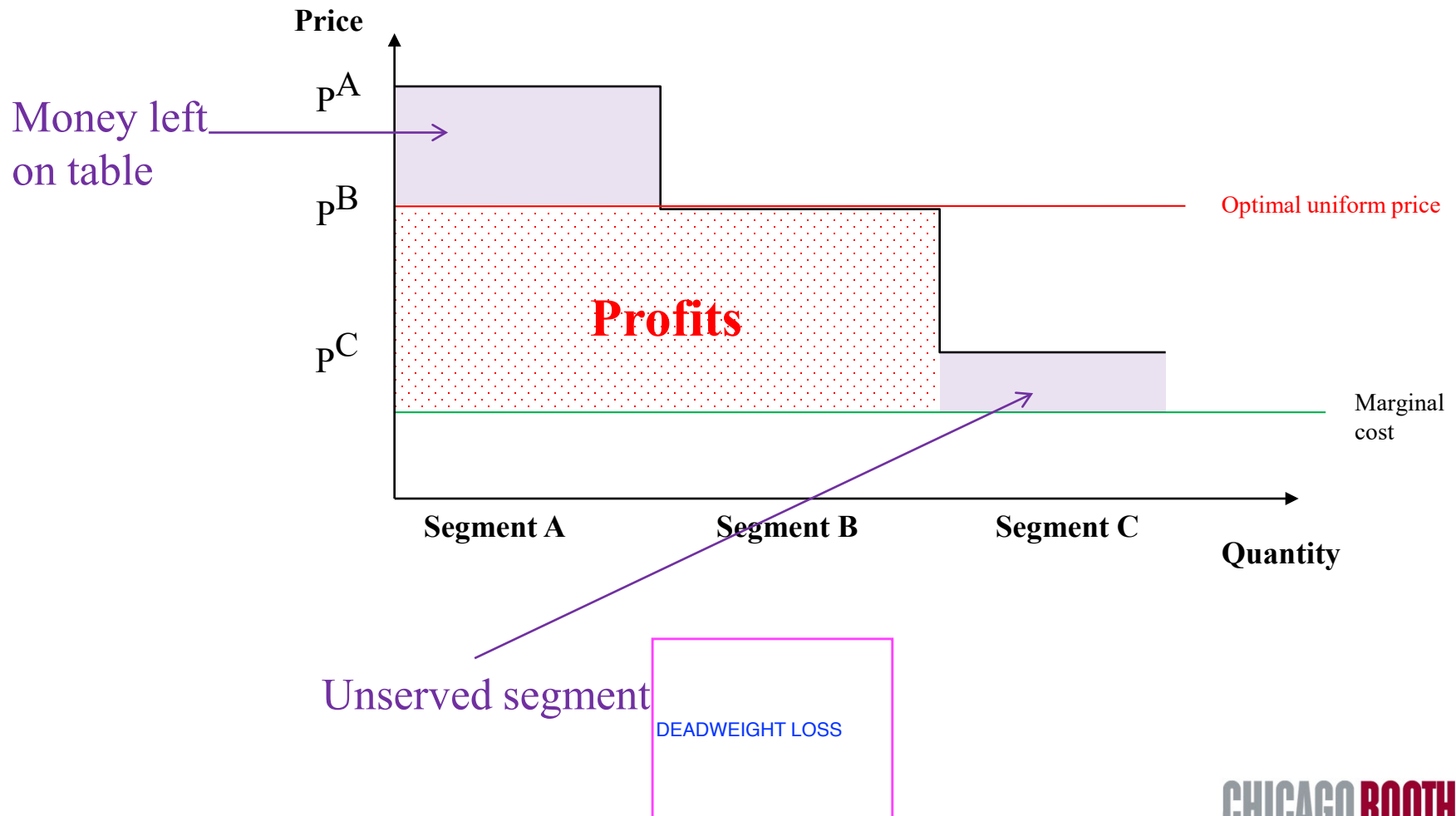
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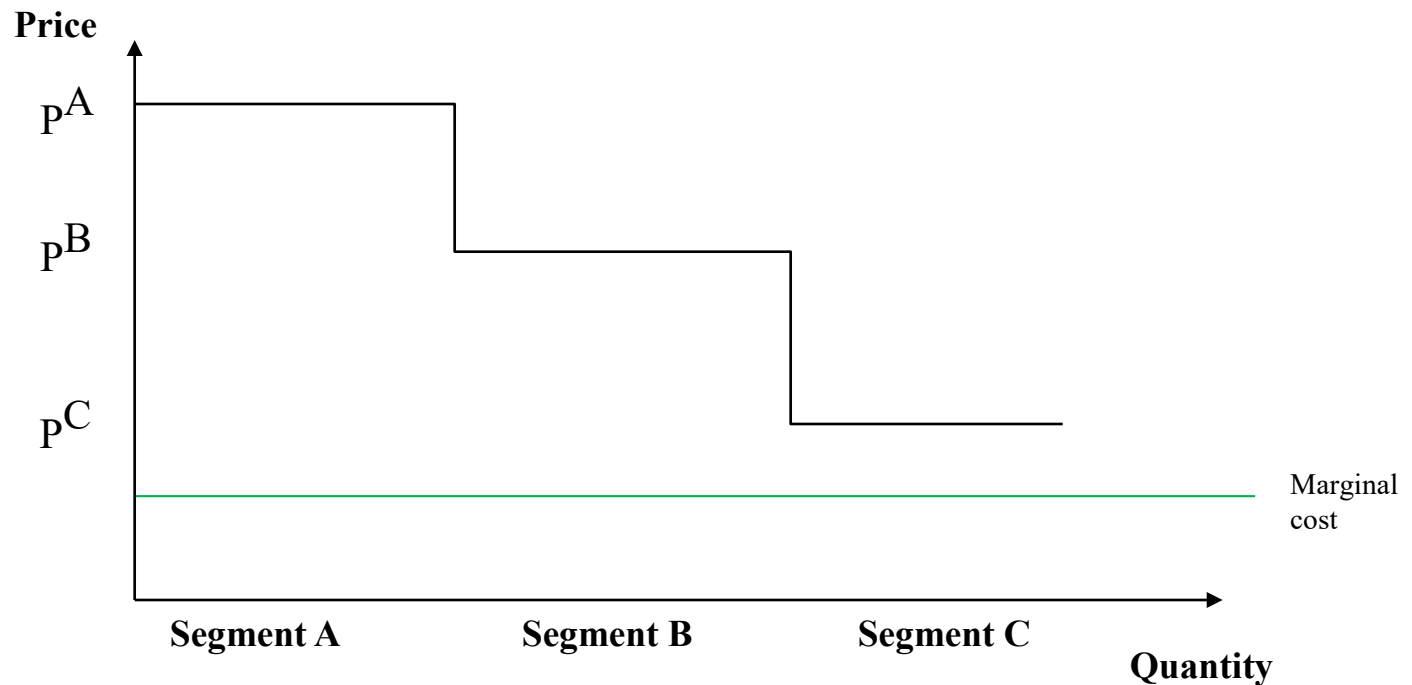
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Targeted Pricing

Suppose we could identify customers/segments

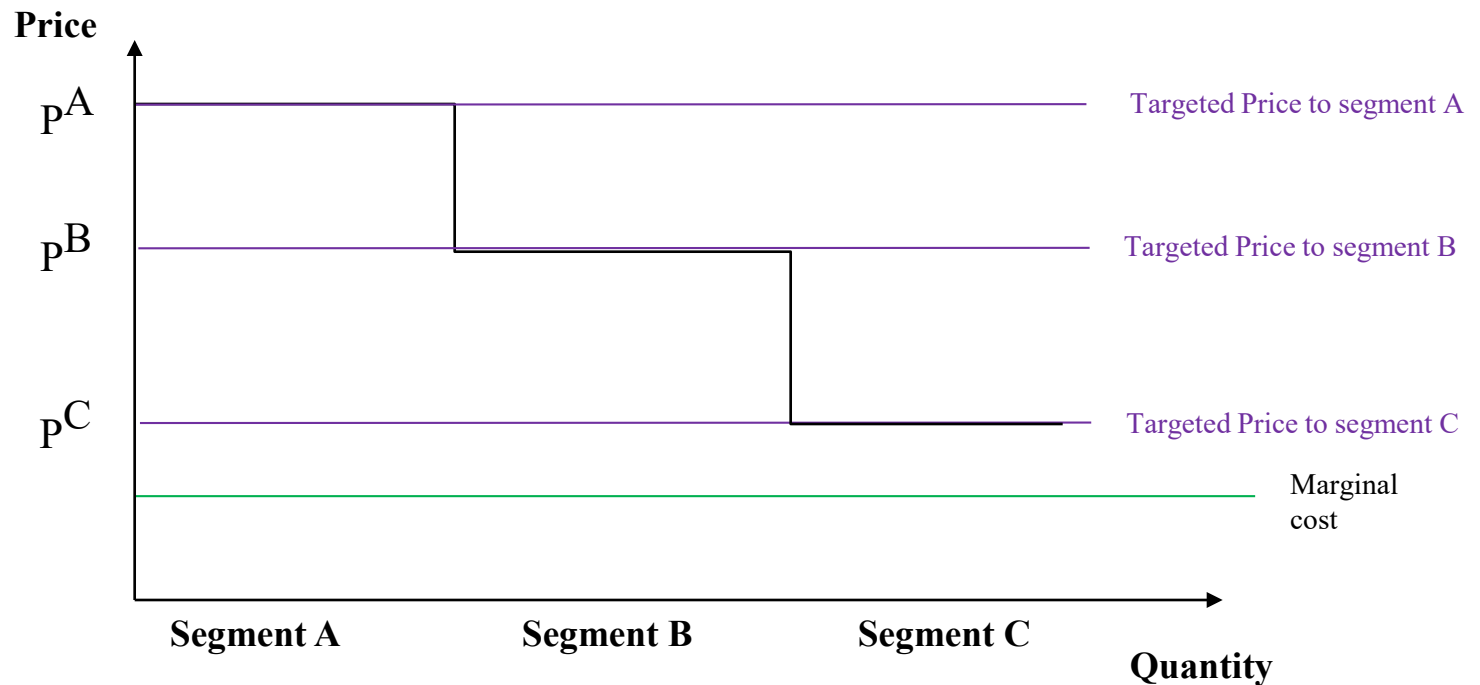
4



Targeted Pricing

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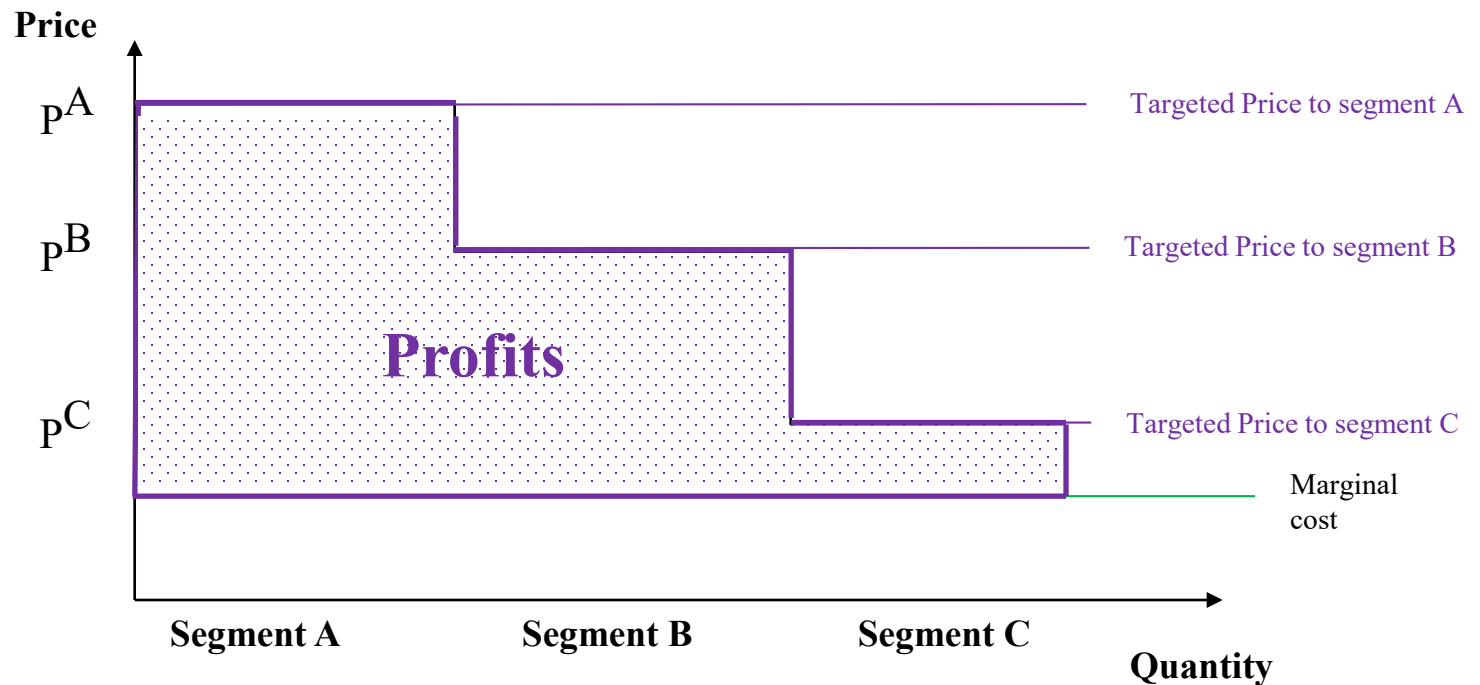
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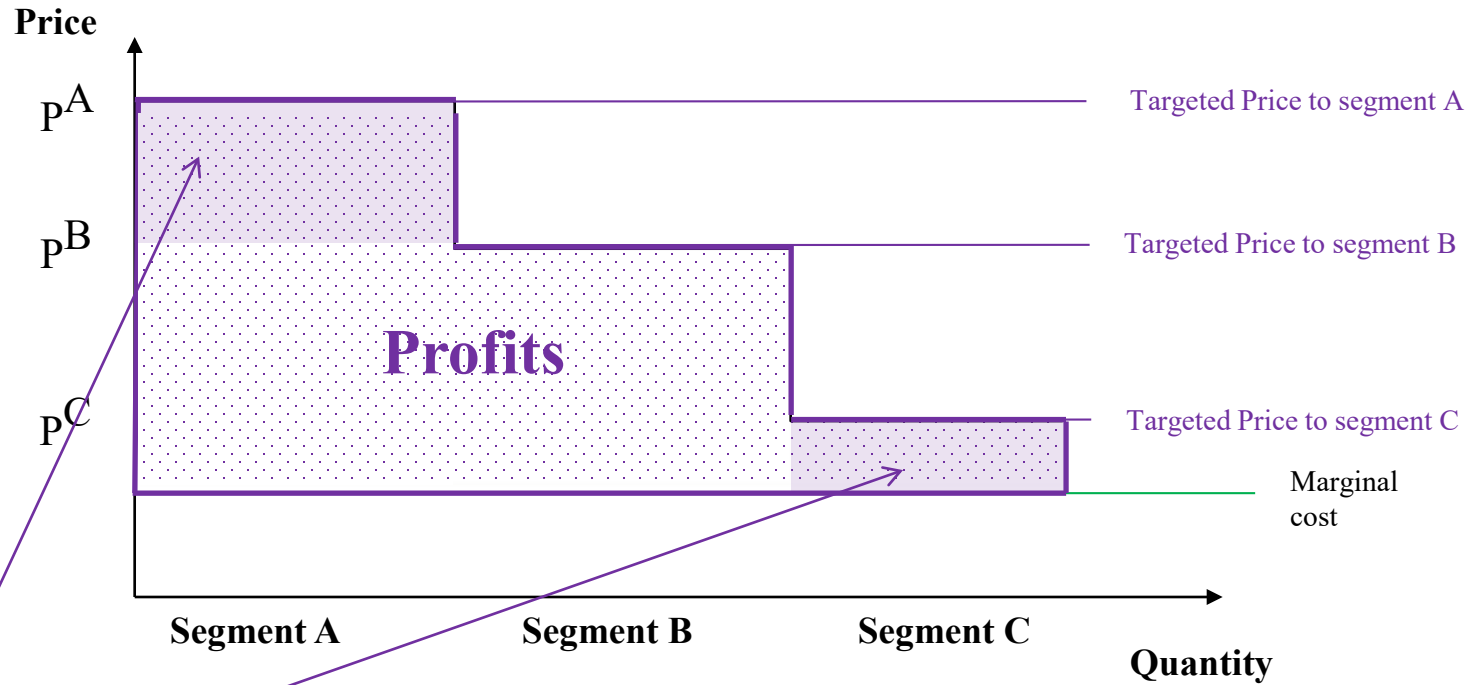
4



Targeted Pricing

Suppose we could identify customers/segments

4



Incremental
Profits
From Targeting

Scalable online price targeting

Example: Ziprecruiter.com

12

- Recall our online recruiting B2B pricing experiment
- Originally charging \$99 per month
- In August 2015, we tested 10 price levels in an experiment
- We determined that \$249 would be much more profitable

Scalable online price targeting

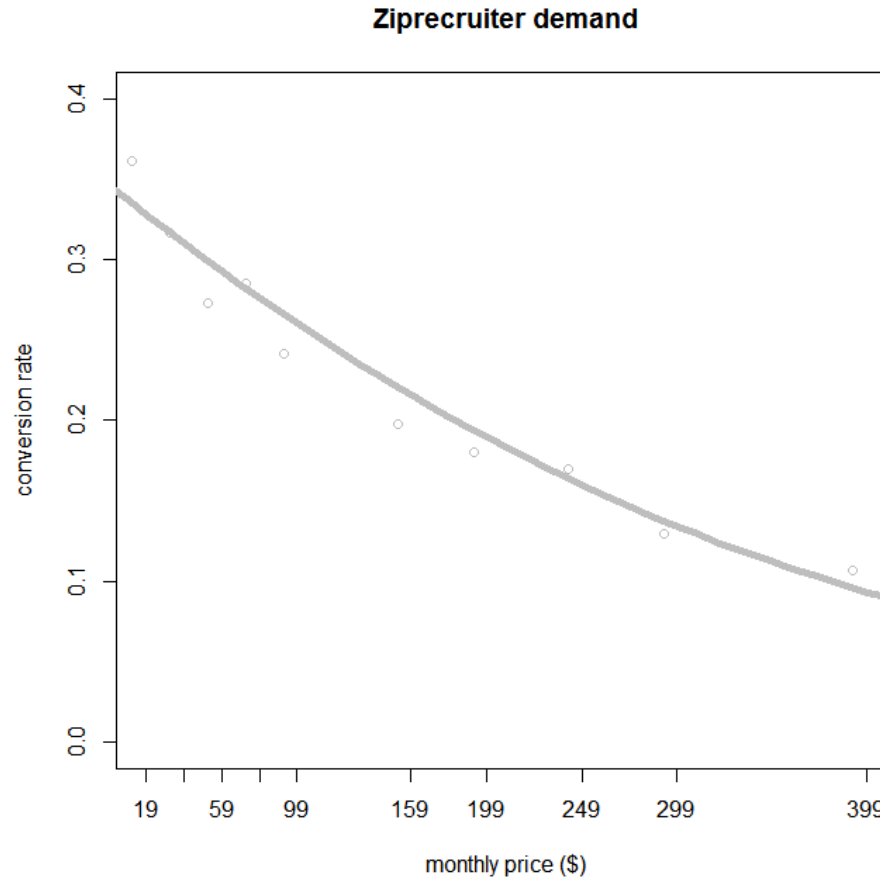
Example: Ziprecruiter.com

13

- Recall our online recruiting B2B pricing experiment
- Originally charging \$99 per month
- In August 2015, we tested 10 price levels in an experiment
- We determined that \$249 would be much more profitable
- In November 2015, we also tested a ***personalized pricing*** strategy using AI

Ziprecruiter Demand

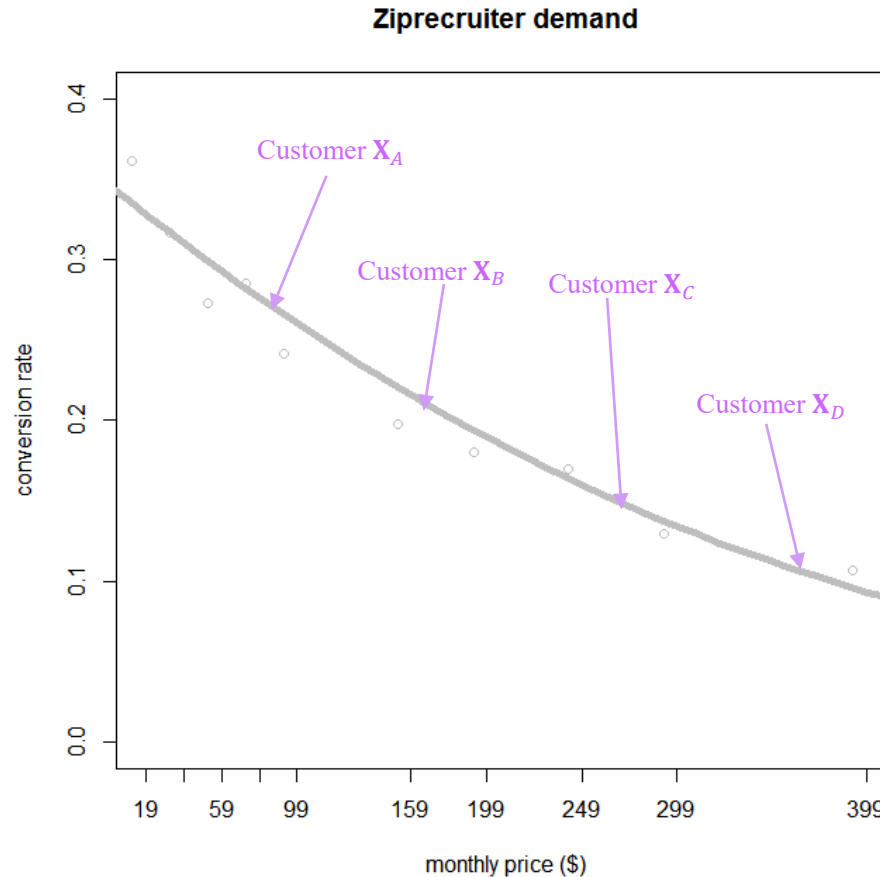
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Ziprecruiter Demand

6

Are there *feature* variables, X , that predict an individual customer's demand before she hits the paywall?



Monetizing Customer Data

7

Customer data Ziprecruiter observes at registration stage

Company/Job Variable Names
Job state
Company type
Has commissions
Company declared job slots needed
Job total benefits
Employment type
Resume required
Job medical benefit
Job vision benefit
Job life insurance benefit
Job category

Monetizing Customer Data

7

Customer data Ziprecruiter observes at registration stage

Company/Job Variable Names
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Job total benefits
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Job category

Can Ziprecruiter **monetize** these data?

Scalable **Personalized** Pricing

Targeting as a **Big Data** problem

8

- Thousands of potentially targetable *features*
 - ▣ As many features as observations!
- Run **Weighted-Bootstrap Lasso Regression**
 - ▣ Combine statistics and machine learning
 - ▣ Link WTP to features



Scalable **Personalized** Pricing

Targeting as a **Big Data** problem

8

- Thousands of potentially targetable *features*
 - ▣ As many features as observations!

AKA, DEEP MACHINE LEARNING ALGORITHM
- Run **Weighted-Bootstrap Lasso Regression**
 - ▣ Combine statistics and machine learning
 - ▣ Link WTP to features
- Can it work?
 - ▣ Suppose 30 “relevant” dummy variables:

$$2^{30} = 1,073,741,824 \text{ segments!}$$

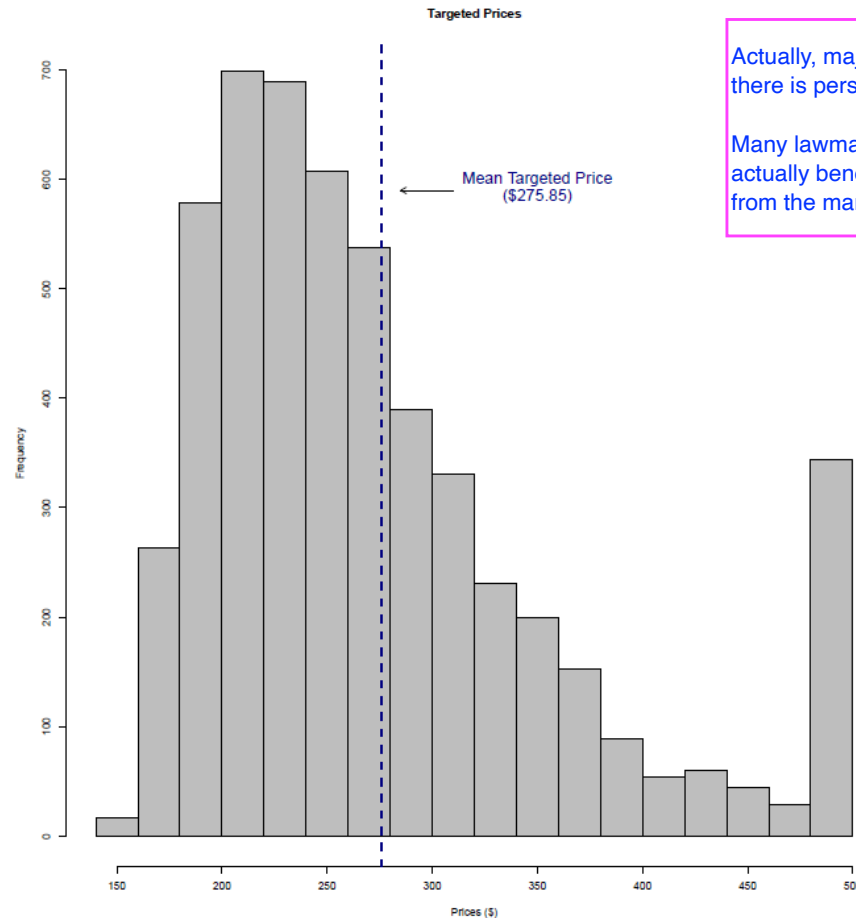


Scalable **Personalized** Pricing

Simulated Personalized Prices for August 2015 training sample

9

apply inverse-elasticity
rule for each customer



Actually, majority of the customers pay below median if there is personalised targeting.

Many lawmakers get this wrong, small customers do actually benefit, because otherwise, they are excluded from the market.

Cap at \$499 to prevent customer backlash or "sticker shock"

Distribution of Personalized Prices (N= 7,867)

Scalable **Personalized** Pricing

Projected Revenues

21



Pricing	Projected Profit Per Lead
Base (\$99)	\$25
Uniform (\$249)	\$40
Targeted	\$44

Scalable **Personalized** Pricing

Validation

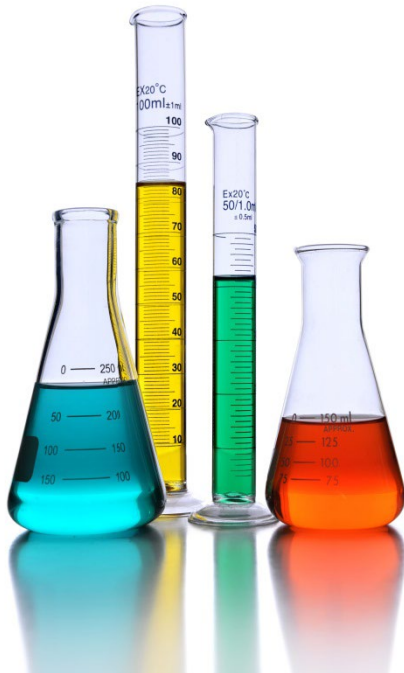
22

Nov 2015: Implement a 3-bin experiment

5,315 new “starter” firms

1. Control pricing – \$99 (25%)
2. Uniform pricing – \$249 (25%)
3. Targeted pricing (50%)

Run and evaluate.



Scalable **Personalized** Pricing

Validation

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Scalable **Personalized** Pricing

Validation

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Pricing	Projected Profit Per Lead	Realized Profit Per Lead
Base (\$99)	\$25	\$23
Uniform (\$249)	\$40	\$38
Targeted	\$44	\$42

Scalable **Personalized** Pricing

Validation

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65%
Rev ↑

Scalable **Personalized** Pricing

Validation

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Scalable **Personalized** Pricing

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83%
Rev ↑

Scalable **Personalized** Pricing

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Uniform (\$249)	\$40	\$38
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83%
Rev ↑

- When we account for statistical error, our results fall within our predictions back in September 2015

→ **A.I.** algorithm seems to work well!!!

Scalable **Personalized** Pricing Evaluation

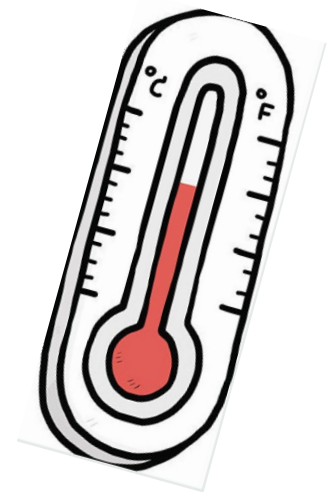


Recall:

Optimization increased monthly profits by 65%

Approximately \$70 million in **incremental**

Scalable **Personalized** Pricing Evaluation



Recall:

Optimization increased monthly profits by 65%

Approximately \$70 million in **incremental**

Personalization could have pushed it over \$80 million

Challenges with Targeted Pricing

Is it Legal?

14

- U.S. law (E.U. is stricter on race, nationality & geography):

*“In certain limited circumstances, **price discrimination** might feature as an aspect of an exclusionary strategy meant to enhance or protect market power. Intervention should be limited to preventing these exclusionary abuses... Absent accompanying unfair or deceptive conduct, **personalized pricing** alone does not raise any of these concerns, and therefore provides no basis for intervention.” (FTC, 2018)*

- **Disparate Treatment** prohibited in housing, credit, employment, public accommodations, & voting based on **protected classes**
 - sex, race/color, religion/creed, nationality, sexual orientation/gender identity, disability status, familial status, veteran status, genetic info
- The Pink Tax Repeal Act (HR 2048): prohibits product manufacturers or service providers from selling substantially similar products at different prices based on gender... If, for example, only difference between 2 products is color, they are substantially similar. (pending in Congress)
- See also
 - Katzman V. Victoria's Secret Catalogue 923 F. Supp. 580 (1996)
 - “Battle of the Prices: Is It Ever Fair to Charge One Sex More?” The Atlantic, 2014.

Challenges with Targeted Pricing

Is it **Legal**? Chinese law is stricter

15

Price Law of the People's Republic of China (Dec. 29, 1997)

“Operators shall not engage in the following unfair pricing practices:

5. When providing the same goods or services, **practicing price discrimination** against other operators with the same trading conditions;”

Regulations of Shenzhen Special Economic Zone on Promoting the Artificial Intelligence Industry (March 8, 2024)

“Any organization or individual engaged in AI research or application shall not conduct any of the following acts:

4. Any organization or individual shall **not discriminate** against users based on race, gender, nationality, ethnic group or religious beliefs;

5. Any organization or individual shall not resort to any acts that harms consumers' rights and interests, **such as price discrimination** and consumption fraud, based on user's habits, preferences or ability to pay;”

Challenges with Targeted Pricing

4 Fairness

16

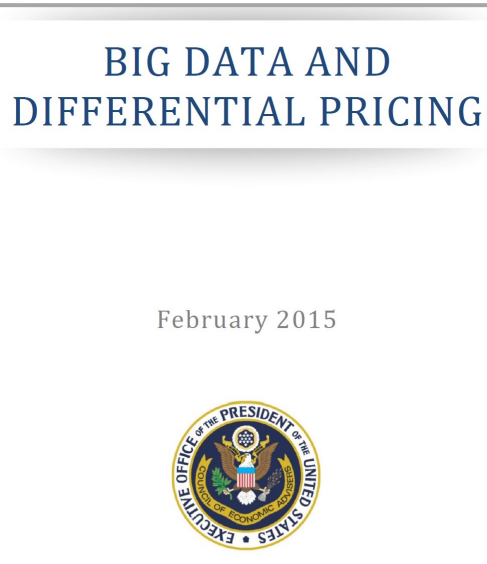
Is targeted pricing **fair**?



Challenges with targeted pricing

Fairness

17

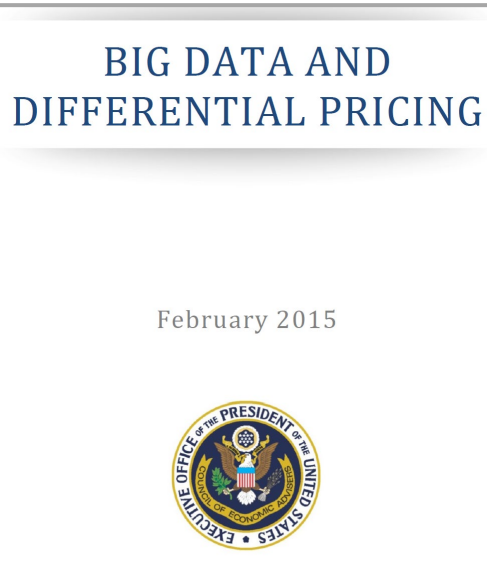


“...[differential pricing] transfers value from consumers to shareholders, which generally leads to an increase in inequality and can therefore be inefficient from a utilitarian standpoint.”

Challenges with targeted pricing

Fairness

17



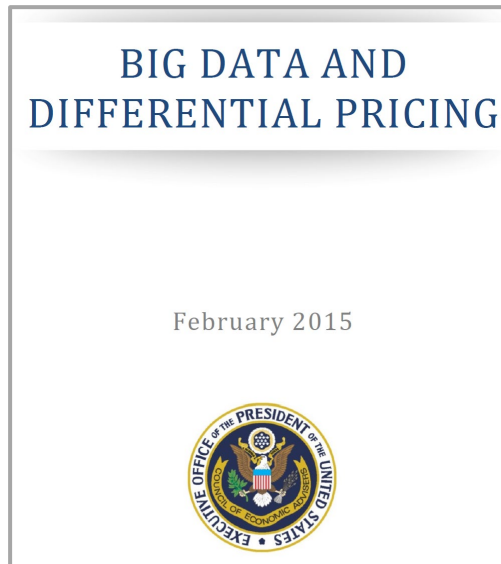
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- Theoretically ... targeting can **increase** total value to customers

Challenges with targeted pricing

Fairness

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“...[differential pricing] transfers value from consumers to shareholders, which generally leads to an increase in inequality and can therefore be inefficient from a utilitarian standpoint.”

- Theoretically ... targeting can **increase** total value to customers

Ziprecruiter: personalized pricing *lowered* prices relative to uniform for 67% of customers!

Challenges with targeted pricing

Fairness & Public Relations Concerns

18

Price transparency can create customer backlash



The Telegraph

Welcome to 'personalised pricing' – where the better off pay more

Chris Stokel-Walker

Sat, June 29, 2024 at 3:00 AM CDT · 6 min read



Firms may avoid targeting for PR reasons



"We've never tested and we never will test prices based on customer demographics"

- Jeff Bezos, CEO, Amazon

□ Also see **extra** slides on other challenges with targeted pricing

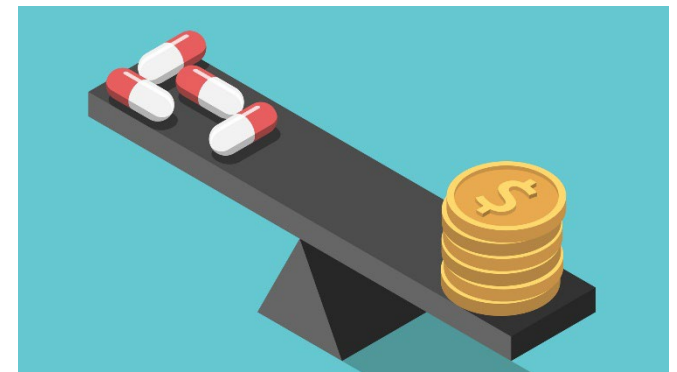
Challenges with targeted pricing

Fairness & International Segmentation

19

“The U.S. subsidizes the worldwide pharmaceutical market. One reason is U.S. prices are higher than elsewhere. If each drug had a single international price across the highest-income OECD countries, and total pharmaceutical firm profits were held fixed, then U.S. prices would fall by half and every other country’s prices would increase (by 28 to 300%). International prices would maintain firms’ R&D incentives and more equitably share the costs of pharmaceutical research.”

[Ho & Pakes \(2024\), National Bureau of Economic Research](#)



Of course, what is “fair”?

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THE WALL STREET JOURNAL



“It’s not about fairness. It’s about the appearance of fairness.”

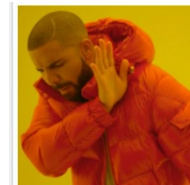
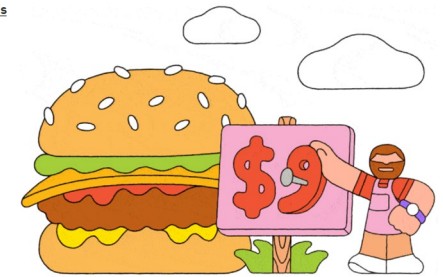
Consumers Hate ‘Price Discrimination,’ but They Sure Love a Discount

The Wendy’s debacle is a warning shot for brands: If you want to play with prices, make sure to communicate why and whom it could help.

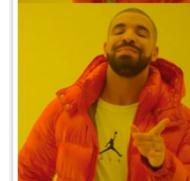


By Lydia DePillis

April 6, 2024



Surge pricing during periods of high demand



Happy hour discount during periods of low demand

Degrees of Price Segmentation

Product Line Pricing or Indirect Pricing

21

- If Targeting is tricky in your product market(s)...
- Suppose you launch different *versions* of your product to charge different prices to different segments.

E.g., high quality for high WTP segments and low quality for low WTP segments...

- *use quality as an indirect “fence” to segment customers*

Indirect vs Direct Price Segmentation

22

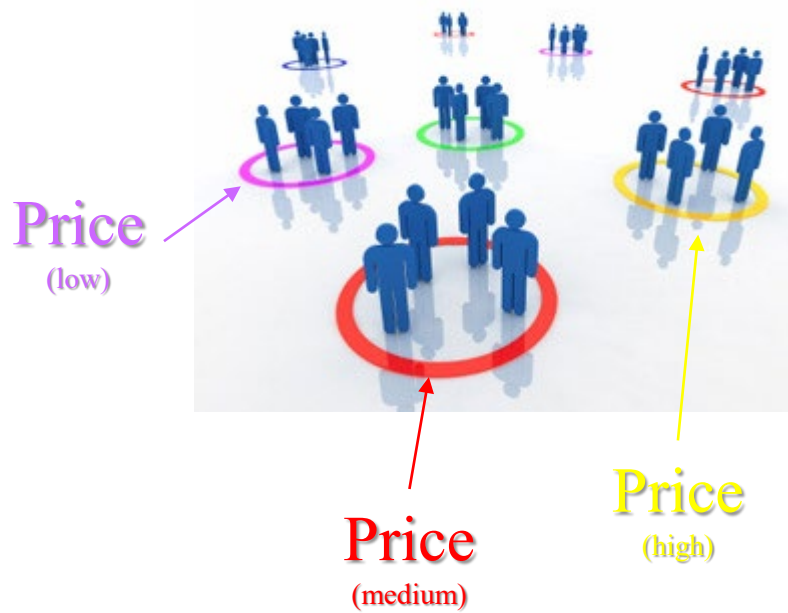
TARGETING



Indirect vs Direct Price Segmentation

22

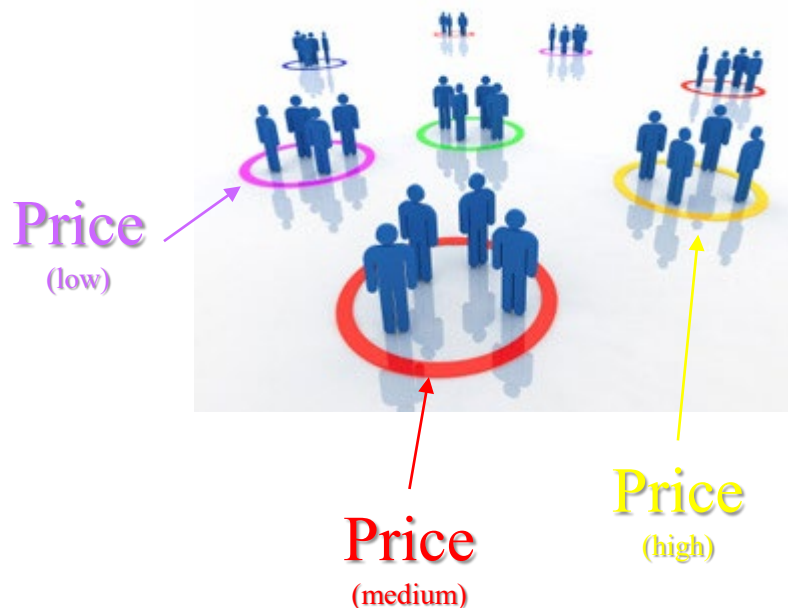
TARGETING



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TARGETING



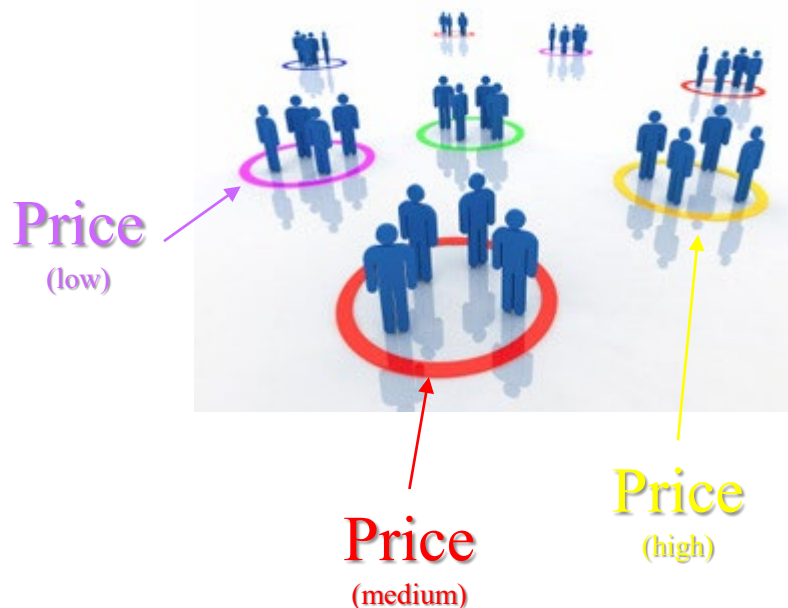
PRODUCT LINE PRICING “versioning”



Indirect vs Direct Price Segmentation

22

TARGETING



PRODUCT LINE PRICING “versioning”



Product 1
Good
Low margin

Product 2
Better
Medium margin

Product 3
Best
High margin

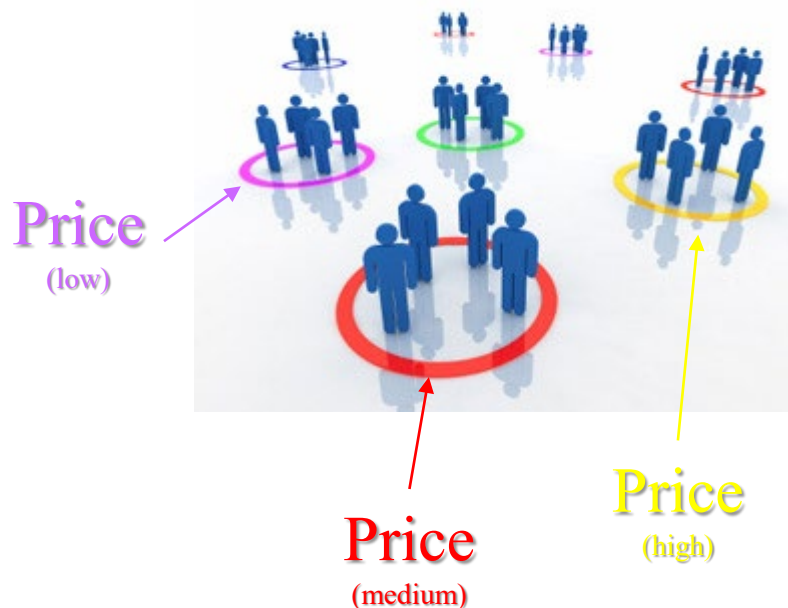
Offer everyone same set of products

Let buyers *self-select*

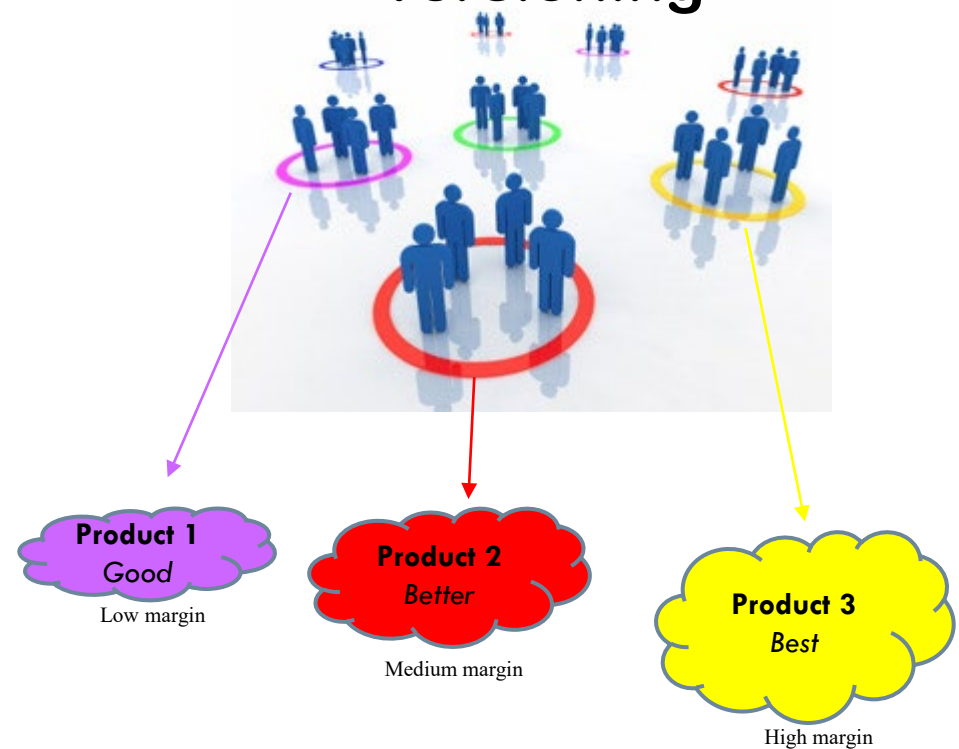
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TARGETING



PRODUCT LINE PRICING “versioning”



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Product Line Pricing Strategy is common in high-tech industries: SaaS

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The screenshot shows the Claris pricing page for Claris Connect. The page has a navigation bar with links: Why Claris, Claris FileMaker, Claris Connect, Resources, Blog, Try, Pricing, and a search icon. The main heading is "Pricing" with a subheading "Find your plan." Below this is a message: "Whether you're a small business owner or belong to a large enterprise, we've got a plan to meet your needs." There are two tabs: "Claris FileMaker" and "Claris Connect", with "Claris Connect" being the active tab. The pricing table has four columns: Developer, FileMaker Plus, Essentials, and Standard. Each column lists the price (per month billed annually), a "Buy" button, and a "Try" button (only for Developer). Below the main table, there are expansion packs for each plan. At the bottom, there is a section "Included with all plans:" with four bullet points. A "See complete feature list" button is at the bottom center, and a chat icon is at the bottom right.

Developer	FileMaker Plus	Essentials	Standard
US\$19⁹⁹	US\$49⁹⁹	US\$99	US\$249
per month billed annually	per month billed annually	per month billed annually	per month billed annually
Buy	Buy	Buy	Buy
Try			
Active flows: No limit	Active flows: No limit	Active flows: 15	Active flows: No limit
API requests: 500 per month	API requests: 10,000 per month	API requests: 10,000 per month	API requests: 50,000 per month
App limit: None	App limit: None	App limit: None	App limit: None
Requires: None	Requires: FileMaker license	Requires: None	Requires: None
Expansion pack — US\$50 Add 5,000 API requests per month.		Expansion pack — US\$100 Add 15,000 API requests per month.	

Included with all plans:

- Access to all standard connectors.
- 30 days of flow history.
- No limit to the number of standard apps and FileMaker apps you can connect.
- Access to all standard features.
- 24/7 support in English. Free for the first year.

[See complete feature list](#)

Platform to consolidate workflow in a company across applications (e.g., automation of processes that need to connect applications)

Also have product for app development.

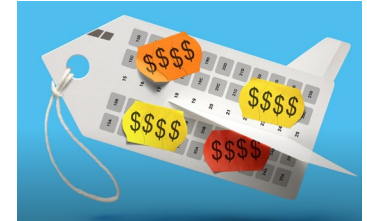
Product Line Pricing Strategy is common in Consumer Industries

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Want to Know Where You're Sitting on Your Next Flight? It'll Cost You

The largest U.S. airlines aren't shy about fees that let you choose regular old seats; European flights can be even tougher

By Dawn Gilbertson, *Wall Street Journal*, April 16, 2024



"American was recently asking an extra \$44 to \$52 one way for a standard seat near the front of the plane for a flight from Dallas to New York this week and \$29 to \$35 for the same flight in June."

AMC Theatres to Change Movie Ticket Prices Based on Seat Location

By Rebecca Rubin

February 6, 2023



Product Line Pricing Strategy

Separation via Indirect Pricing

25

- Suppose we manage a product line with several substitute choices.
 - Customers face a trade-off between quality and price

Product Line Pricing Strategy

Separation via Indirect Pricing

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- Suppose we manage a product line with several substitute choices.
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- **Goal:** segment customers based on what they buy and, hence, what they pay
 - ▣ High WTP selects higher price offer
 - ▣ Low WTP selects lower price offer

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- **Problem:** high WTP customers may “trade down” (*cannibalization*)

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- **Solution:** “Sorting Strategies”
 - ▣ Make the low price offer less attractive
 - ▣ Make the high price offer more attractive

Product Line Pricing Strategy

Separation via Indirect Pricing

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Coordination of **P**rice and **P**roduct

Product Line Pricing Strategy

Separation via Indirect Pricing

26

PRODUCT LINE PRICING “versioning”



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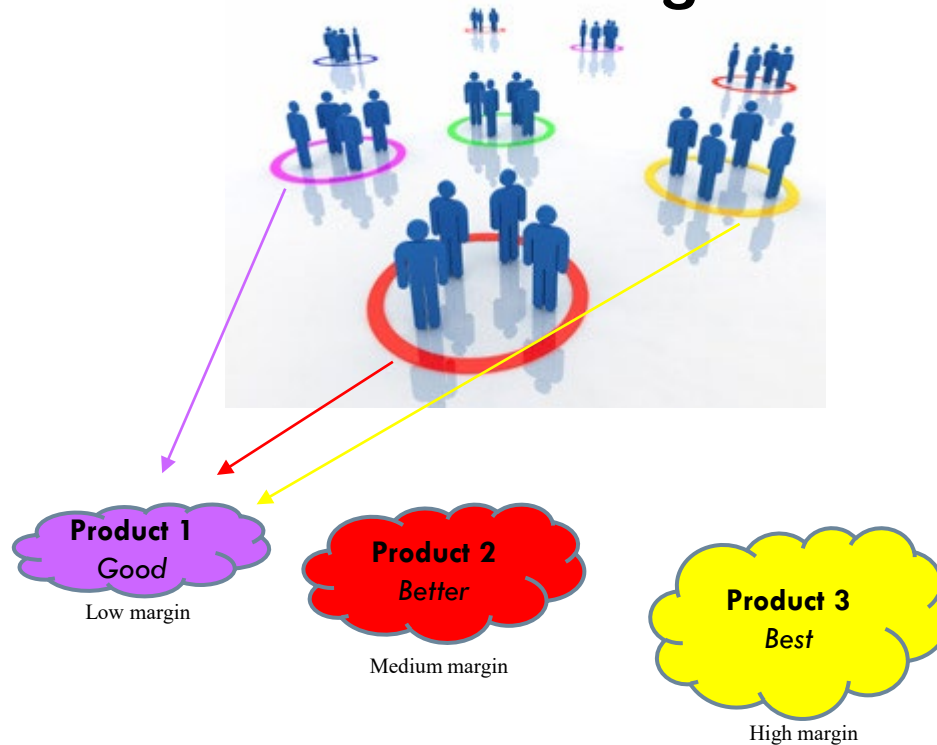
Offer everyone same set of products
Let buyers *self-select*

Product Line Pricing Strategy

Separation via Indirect Pricing

26

PRODUCT LINE PRICING “versioning”



What if low prices draw everyone to low-end product?

Offer everyone same set of products
Let buyers *self-select*

Potential Cannibalization at McDonald's

Example: The McDouble

27

- McDouble: \$1.00
 - ▣ Permanent dollar-menu item since 2008 to compete with other Restaurants



Potential Cannibalization at McDonald's

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*“...the **McDonald's McDouble**, at 390 Calories, 23g (half a daily serving) of protein, 7% of daily fiber, 20% of daily calcium and iron, etc., is the cheapest, most nutritious, and bountiful food that has ever existed in human history.”*

Freakonomics Blog

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Freakonomics Blog



- Big Mac costs about \$3.99
 - ▣ Cannibalization concerns?
 - ▣ Differentiation value of Big Mac vs McDouble?

Same store sales was declining, but the \$1 menu was growing.

So how much of the growth (of Mcdouble) is coming from Big Mac?

Managing a Product Line

If trading down occurs, ...

28

- **P**rice changes
 - ▣ Lower P of high end offer
 - ▣ Raise P of low end offer

- **P**roduct Value Changes
 - ▣ Increase attractiveness of high end offer
 - ▣ Decrease attractiveness of low end offer

See “[Why Some Companies Make Their Products Worse](#)” Richard Davies, *Economist*, 9-15-2016

and “[How did Tesla make some of its cars travel further during Hurricane Irma?](#)” *The Guardian*, 9-17-2017

Elon musk tweet. 100 more miles. so actually the base model S has same range as the high end model. the BMC controls the limited range.

Now, still ongoing class action lawsuit; who actually owns the range?

Car Stereo Example

Optional stereo upgrade in a BMW i4

29



Default Audio

10 speakers

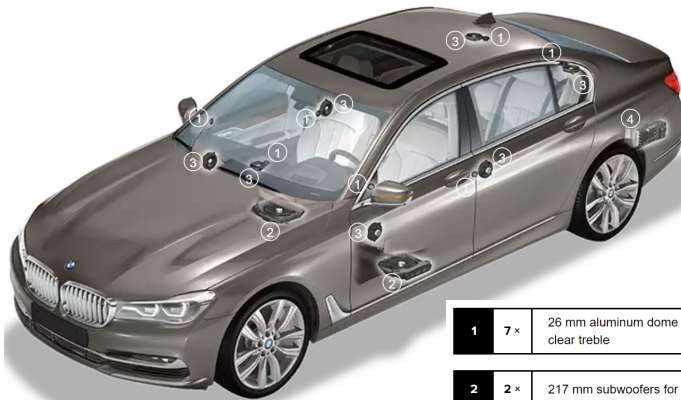
205 Watts

Optional Upgrade

multi-channel Harman Kardon
surround

17 speakers

464 Watts



1 7 x 26 mm aluminum dome tweeters for crystal clear treble

2 2 x 217 mm subwoofers for a punchy bass

3 7 x 100 mm midrange speakers for clear and accurate reproduction of voices

4 464 watt [Logic 7](#) amplifier for a dynamic listening experience

*Representative vehicle model shown. Speaker specifications & locations may vary.

Assume: cost of stereo to dealer is \$0 (i.e., sunk) to simplify calculations.

Car Stereo Example

Optional stereo upgrade in a BMW i4

30

- Value of base car
 - High consumer Value segment (50%): \$75,000
 - Low consumer Value segment (50%): \$55,000
- What should we charge for a car?



Marketing
research

Car Stereo Example

Optional stereo upgrade in a BMW i4

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□ Value of base car

- High consumer Value segment (50%): \$75,000
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Marketing
research

□ What should we charge for a car?

	Upgrade Valuations	
Segment	Harman Kardon	Generic
High segment values at ...	\$ 3,000	\$ 250
Low Segment values at ...	\$ 500	\$ 100

Marketing
research

Car Stereo Example

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Marketing
research

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Marketing
research

□ Product line:

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Marketing
research

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	Upgrade Valuations	
Segment	Harman Kardon	Generic
High segment values at ...	\$ 3,000	\$ 250
Low Segment values at ...	\$ 500	\$ 100

Marketing
research

□ Product line:

- 1 car with Generic stereo; and
- 2 car with Harman Kardon stereo upgrade

Car Stereo Example

Optional stereo upgrade in a BMW i4

32

□ Use EVC

□ $Price^{Gen} = \$55,100$

Segment	Harman Kardon	Generic
High	\$ 3,000	\$ 250
Low	\$ 500	\$ 100

Car Stereo Example

Optional stereo upgrade in a BMW i4

32

- Use EVC

- $Price^{Gen} = \$55,100$
 $WTP^{HK} = DV + RV$

Segment	Harman Kardon	Generic
High	\$ 3,000	\$ 250
Low	\$ 500	\$ 100

Car Stereo Example

Optional stereo upgrade in a BMW i4

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□ Use EVC

□ $Price^{Gen} = \$55,100$

$$WTP^{HK} = DV + RV$$

$$= (\$3,000 - \$250) + \$55,100$$

Segment	Harman Kardon	Generic
High	\$ 3,000	\$ 250
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Car Stereo Example

Optional stereo upgrade in a BMW i4

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□ Use EVC

□ $Price^{Gen} = \$55,100$

$$WTP^{HK} = DV + RV$$

$$= (\$3,000 - \$250) + \$55,100$$

$$= \$57,850$$

Segment	Harman Kardon	Generic
High	\$ 3,000	\$ 250
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Car Stereo Example

Optional stereo upgrade in a BMW i4

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$$= \$57,850$$

Segment	Harman Kardon	Generic
High	\$ 3,000	\$ 250
Low	\$ 500	\$ 100

➤ $Price^{HK} = \$57,850$

Car Stereo Example

Optional stereo upgrade in a BMW i4

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□ Use EVC

$$\square \text{ Price}^{Gen} = \$55,100$$

$$WTP^{HK} = DV + RV$$

$$= (\$3,000 - \$250) + \$55,100 \\ = \$57,850$$

High WTP customers willing to pay \$3000-\$250 for the high end car (the difference between how they are willing to pay for HK above Generic)

If the reference product is appealing, it is very hard to sell a high end product at a high premium.

Segment	Harman Kardon	Generic
High	\$ 3,000	\$ 250
Low	\$ 500	\$ 100

$$\triangleright \text{Price}^{HK} = \$57,850$$

What can i monetize? I cant monetize the full use value. i can monetize the economic value.
once we sell the Generic model, it creates cross price elasticity which affects my ability to maximise the price for HK cars.

$$\square \text{ Revenue} = \$55,100 + \$57,850 = \$112,950 \text{ (vs. \$110,000)}$$

Differential pricing gain extra \$2950 (not alot compared to Ziprecruiter)

indirect segmenting via product line choices.

The whole point of product line is to create differentiation and reduce cannibalisation.

Car Stereo Example

Optional stereo upgrade in a BMW i4

33

- Exercise: What is the optimal product line pricing structure if data results for stereo preferences had turned out as follows:

	Upgrade Valuations	
Segment	Harman Kardon	Generic
High segment values at ...	\$ 3,000	\$ 2,500
Low Segment values at ...	\$ 500	\$ 100

Car Stereo Example

Price Coordination

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- Car stereo example highlights role of **P**rice coordination in product line pricing
 - ▣ You do not capture all the value on Harman Kardon
 - ▣ Otherwise, Generic stereo cannibalizes Harman Kardon

- So product line pricing **tactics** also need to be coordinated

Product Line Management

Coordination of brand manager incentives (**HR**)

35

- Most firms allocate brand-level decision-making to a brand management team with P&L responsibility only for its brand(s)
- *What incentives does a brand manager have to coordinate her pricing decisions with substitute goods managed by another manager in the same company?*
- **Ignoring cannibalization can create competition between products or divisions within the same firm!**

Example; the \$1 menu on McDonalds. That is wrongful cannibalisation internally amongst department (\$1 department and normal team).

Take Away

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- **Targeted Pricing** is huge opportunity if you sell to very heterogeneous customers. But, it is challenging to implement.
- **Product Line Pricing** is ultimately about coordination and managing customers' incentives:
 - serve low-end segments; but
 - deter high-end segments from *trading-down*
- Uncoordinated tactical pricing → compete with self!
- **Psychology**: Can there be too much variety? See Appendix for **Choice Overload**.

PLP technique feels more ethical, but risks cannibalisation if done wrongly.

Victoria's Not-So-Secret Price Discrimination

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“Sometimes price discrimination can end up being costly not just to consumers, but also to producers. In 1996 Denise Katzman of New York City sued Victoria's Secret for gender discrimination and ask for millions of dollars in damages. In alleging gender discrimination, Katzman didn't object to the catalog's pages of scantily clad women. Rather, she pointed to the promotional coupon on the catalog's back page.

The problem? While Ms. Katzman's catalog offered her a \$10 off an order of \$74, an almost identical catalog for a male friend offered \$25 off the same amount. Was her catalog out of date?

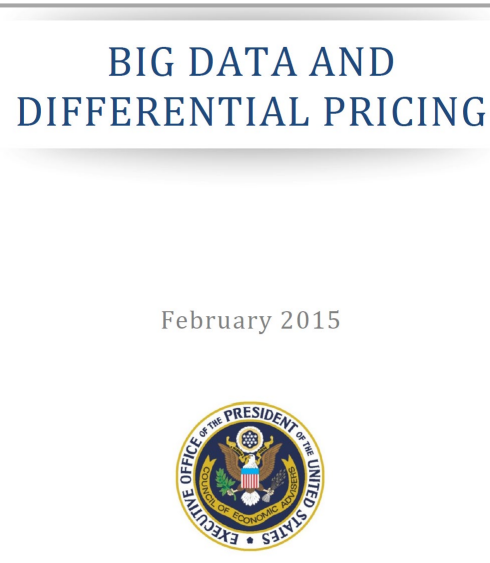
Nope. The folks at Victoria's Secret were just engaging in a little “naked” price discrimination.”



Challenges with Targeted Pricing

Fairness & Marketing for Good

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“...the combination of differential pricing and big data raises concerns that some consumers can be made worse off, and have very little knowledge why.”

“...[differential pricing] transfers value from consumers to shareholders, which generally leads to an increase in inequality and can therefore be inefficient from a utilitarian standpoint.”

How should we define *Price Fairness*?

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Uniform offers versus uniform access/affordability?

- Personalized Pricing at Ziprecruiter:

Over 60% customers pay less under personalized pricing (small firms)!

Source: [Dubé and Misra \(2022\)](#)

- Uniform pricing in US grocery/drug/convenience:

Prices in poor neighborhoods 8% higher than if targeted geographically.

Source: [DellaVigna and Gentzkow \(2017\)](#)

- Personalized Pricing & Nutrition:

Low-income consumers have lower WTP for nutritious foods.

Source: [Allcott et al \(2019\)](#), [Dubé \(2020\)](#)

- Personalized Municipal fines/fees:

Escalating fines/fees increase bankruptcies & evictions in minority neighborhoods.

Source: [Dubé \(2021\)](#), [Glenn et al. \(2022\)](#)

- Pfizer to Sell Vaccines, Drugs at Low Prices to Poorer Countries

...the drugmaker will sell 23 products at not-for-profit prices to 45 countries

Source: [Wall Street Journal, May 26, 2022](#)



"It's not about fairness. It's about
the appearance of fairness."

GDPR and Privacy



GDPR is vague

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- Requires that use of personal data allows for provision:

“meaningful information about the logic involved, as well as the significance and the envisaged consequences of such processing for the data subject.”

- Article 13(2f) and 14(2g) GDPR clarifies:

“The controller should find simple ways to tell the data subject about the rationale behind, or the criteria relied on in reaching the decision. The GDPR requires the controller to provide meaningful information about the logic involved, not necessarily a complex explanation of the algorithms used or disclosure of the full algorithm. The information provided should, however, be sufficiently comprehensive for the data subject to understand the reasons for the decision”

Appendix: Choice Overload

Choice Overload

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- Economic theory suggested that *more* items in the product line can improve profits through segmentation...
- Is it possible to have too many items on the choice board?
- Psychologically, can there be “too much” variety?
- Does this adversely affect your pricing power?

Choice Overload

Assortment at McDonalds

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Burgers



Big Mac



Quarter Pounder with Cheese



Bacon Clubhouse Burger



Bacon Habanero Ranch Quarter Pounder



Bacon & Cheese Quarter Pounder



Deluxe Quarter Pounder



Double Quarter Pounder with Cheese



Hamburger



Cheeseburger



BBQ Ranch Burger



Grilled Onion Cheddar Burger



Double Cheeseburger



McDouble



Jalapeño Double



Bacon McDouble



Daily Double



Premium Crispy Chicken Bacon Clubhouse Sandwich



Premium Grilled Chicken Bacon Clubhouse Sandwich



Premium Crispy Chicken Classic Sandwich



Premium Grilled Chicken Classic Sandwich



Premium Crispy Chicken Club Sandwich



Premium Grilled Chicken Club Sandwich



Premium Crispy Chicken Ranch BLT Sandwich



Premium Grilled Chicken Ranch BLT Sandwich



McChicken



Buffalo Ranch McChicken



Bacon Buffalo Ranch McChicken



Bacon Cheddar McChicken



Southern Style Crispy Chicken Sandwich



Filet-O-Fish



McRib

Choice Overload

Assortment at McDonalds

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Burgers



segmentation
vs
overload?

Choice Overload

Assortment at McDonalds

46

Burgers



segmentation
vs
overload?

“McDonald’s Is Ditching a Bunch
Of Menu Items To Better Its
Customer Experience,”

Business2community, by James Kosur, 1-23-2015

Choice Overload

The iPhone 12 product line

47



“Shopping for an iPhone now requires a magnifying glass for careful inspection of the features—notably the cameras, screen sizes and compatibility with the new 5G cellular networks. It’s important to understand all your choices—what you get and don’t get with each of them, new and old.”

Joanna Stern, “iPhone 12 vs. iPhone 11 vs. iPhone XR vs. iPhone SE: Comparing Apple’s Full Lineup,” *WSJ*, 10-14-2020

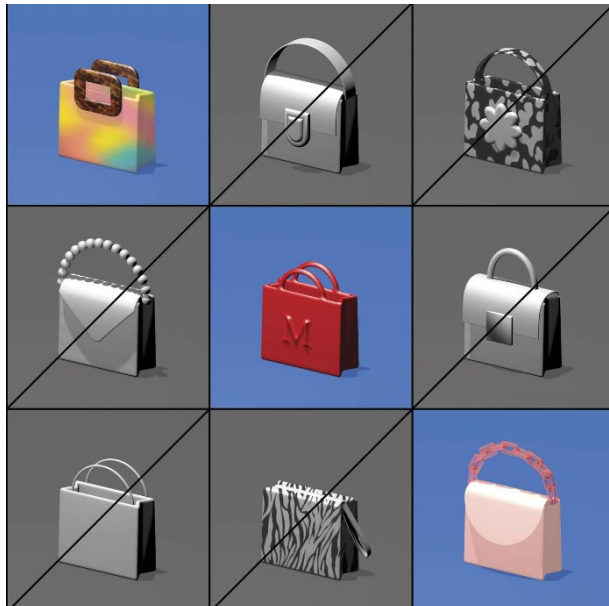
Choice Overload

SKU-rationalization in retail

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Retailers Cut Back on Choices; ‘We Don’t Need Three Types of Red’
Bed Bath & Beyond, Kohl’s and others cull styles to combat decision paralysis, unclog supply chains and reduce end-of-season markdowns

Suzanne Kapner, *WSJ*, 11-22-2020



Offering 1,000 handbag models in a rainbow of colors used to lead to end-of-season discounting for Coach, which has halved the number of designs it offers, focused on three popular hues.

Choice Overload Theory

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- Research in customer psychology finds cases where too many options make customers worse off
 - ▣ Purchase decisions are more complicated
 - ▣ More potential for ex post regret
 - ▣ Customer may decide not to buy

Testing Choice Overload

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THE JAM STUDY

A grocery store conducted 2 tasting sessions. In one session shoppers were allowed to sample 24 flavors of jams, and in the other session they were allowed to sample 6 flavors



24 Choices of Jam **vs** 6 Choices of Jam

Testing Choice Overload

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THE JAM STUDY

A grocery store conducted 2 tasting sessions. In one session shoppers were allowed to sample 24 flavors of jams, and in the other session they were allowed to sample 6 flavors



24 Choices of Jam	vs	6 Choices of Jam
Attracted 60% of Shoppers		Attracted 40% of Shoppers

Testing Choice Overload

50

THE JAM STUDY

A grocery store conducted 2 tasting sessions. In one session shoppers were allowed to sample 24 flavors of jams, and in the other session they were allowed to sample 6 flavors



24 Choices of Jam vs **6 Choices of Jam**

Attracted **60%**
of Shoppers

Shoppers sampled **2**
flavours on average

Attracted **40%** of
Shoppers

Shoppers sampled **2**
flavours on average

Testing Choice Overload

50

THE JAM STUDY

A grocery store conducted 2 tasting sessions. In one session shoppers were allowed to sample 24 flavors of jams, and in the other session they were allowed to sample 6 flavors



24 Choices of Jam vs **6 Choices of Jam**

Attracted **60%**
of Shoppers

Shoppers sampled **2**
flavours on average

3% of shoppers
bought jam

Attracted **40%** of
Shoppers

Shoppers sampled **2**
flavours on average

30% of shoppers
bought jam

Testing Choice Overload

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THE JAM STUDY

A grocery store conducted 2 tasting sessions. In one session shoppers were allowed to sample 24 flavors of jams, and in the other session they were allowed to sample 6 flavors



24 Choices of Jam vs 6 Choices of Jam

Attracted **60%**
of Shoppers

Shoppers sampled **2**
flavours on average

3% of shoppers
bought jam

Conversion: 1.8%

Attracted **40%** of
Shoppers

Shoppers sampled **2**
flavours on average

30% of shoppers
bought jam

Conversion: 12%